

SCI for OTel Assembly

Orientation - July 2026



Agenda

- Software Carbon Intensity for OpenTelemetry
- Assembly Experience
- Consensus in the GSF
- Humans-in-the-Loop

Team



Russell Trow
Operations Director
Green Software Foundation

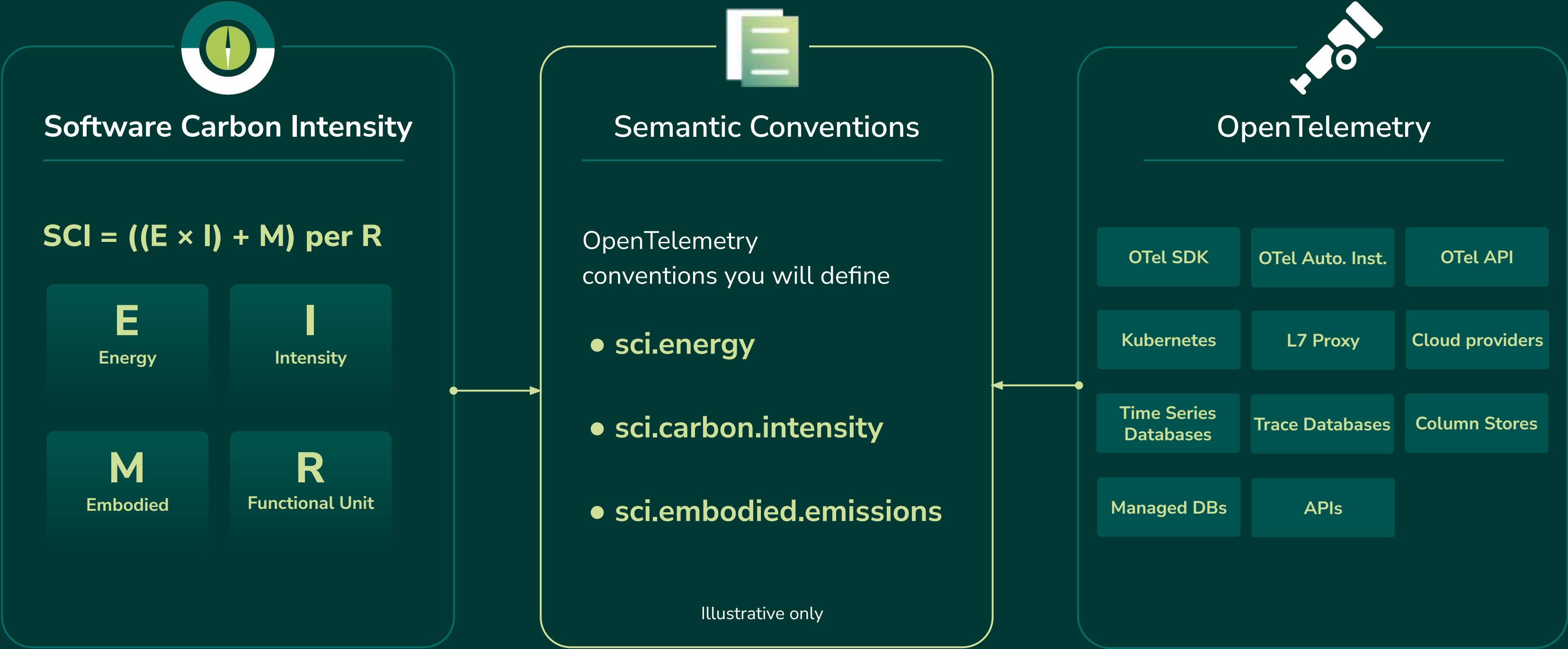


Sarah Hsu
Site Reliability Engineer
Goldman Sachs

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practitioners, maintainers and researchers from
the OpenTelemetry, energy & green software
communities

Software Carbon Intensity in OpenTelemetry



Software Carbon Intensity (SCI)

$$SCI = ((E \times I) + M) \text{ per } R$$

E

Energy

Energy consumed
by the software

I

Intensity

Carbon intensity of
the electricity grid

M

Embodied

Emissions baked into
the hardware

R

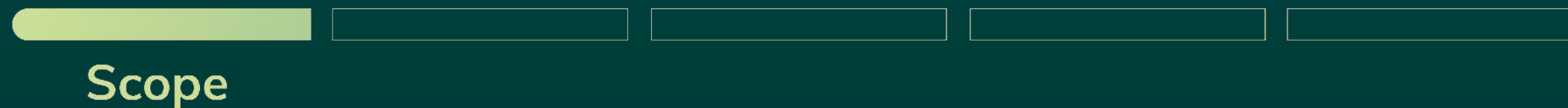
Functional Unit

Rate — per
transaction, user, request

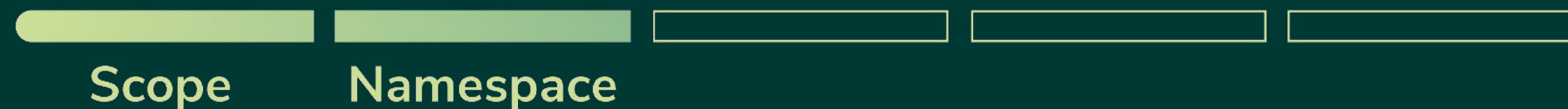
Operational Energy

ISO/IEC 21031:2024 | Designed for Action not Reporting

$$SCI = ((E * I) + M) \text{ per } R$$



$$SCI = ((E * I) + M) \text{ per } R$$



E x I: Operational Energy

$$SCI = ((E * I) + M) \text{ per } R$$



M: Embodied Carbon

$$SCI = ((E * I) + M) \text{ per } R$$



$$SCI = ((E * I) + M) \text{ per } R$$





The Assembly Experience

DISCOVER

Divergence.
What does everyone think?

Answer Questions



DELIBERATE

Convergence.
Which approach is the best?

Rate Candidates



DECIDE

Consensus.
Can we all live with this?

Endorse / Consent / Object



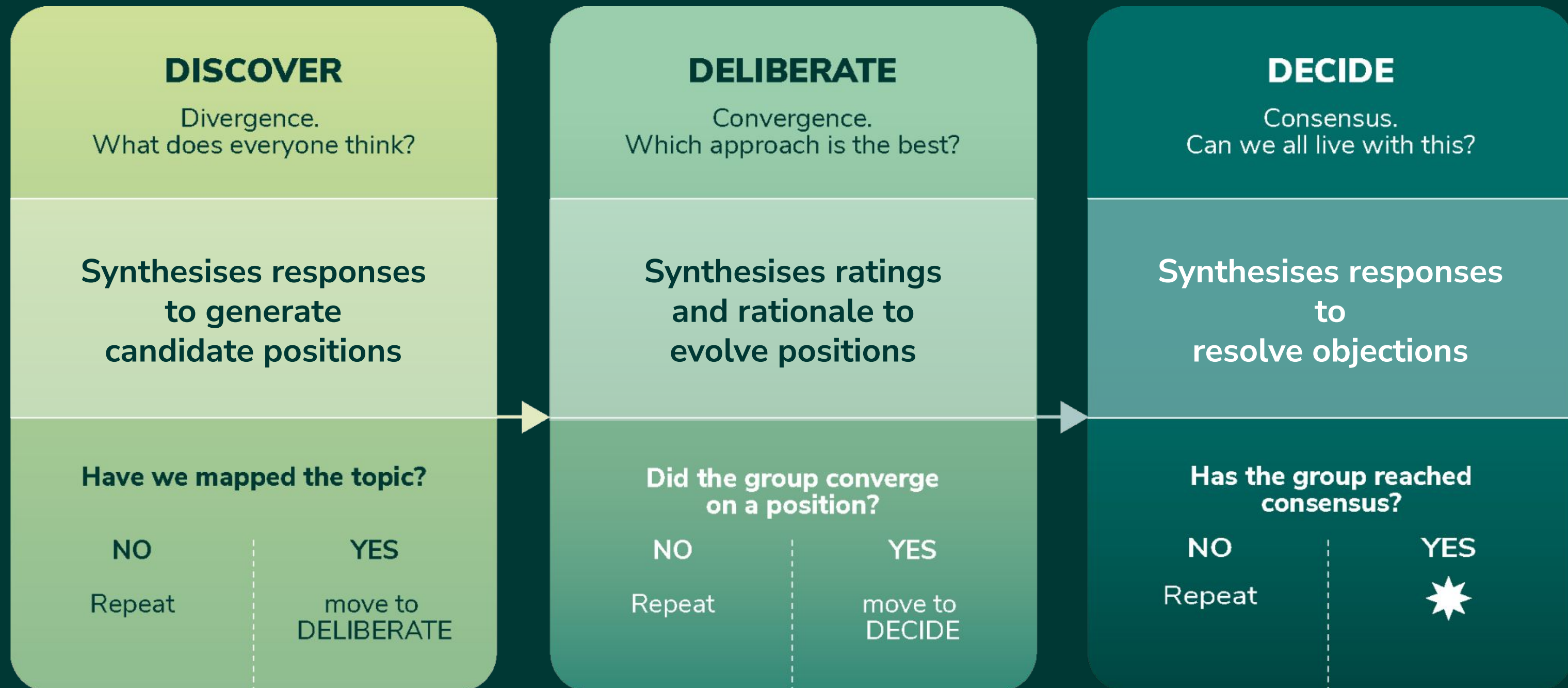
Scope

Namespace

E*I

M

R



Scope

Namespace

E*I

M

R

What We need from You



Respond on Time

A late reply is a missed one



Be direct

Your position, not a wall of caveats



Stay on topic

Answer what the round asks



Respond honestly

Say what you mean, especially if you disagree



Stay Committed

This is lighter than a weekly meeting, but it is not nothing.

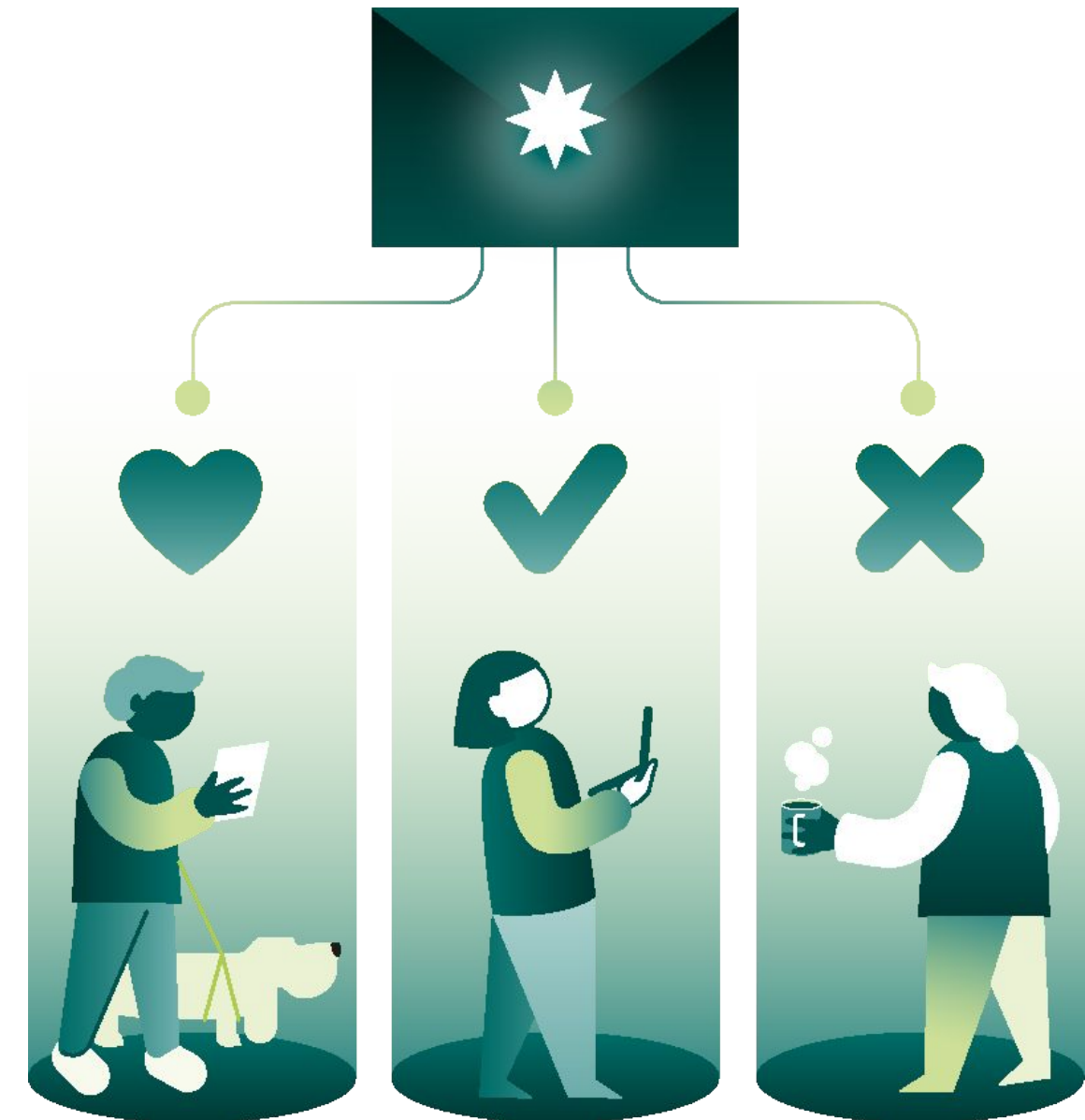
Decision

ENDORSE: Strongly agree & actively support this position

CONSENT: Agree with the position, potentially with reservations

OBJECT: Do not agree to the position

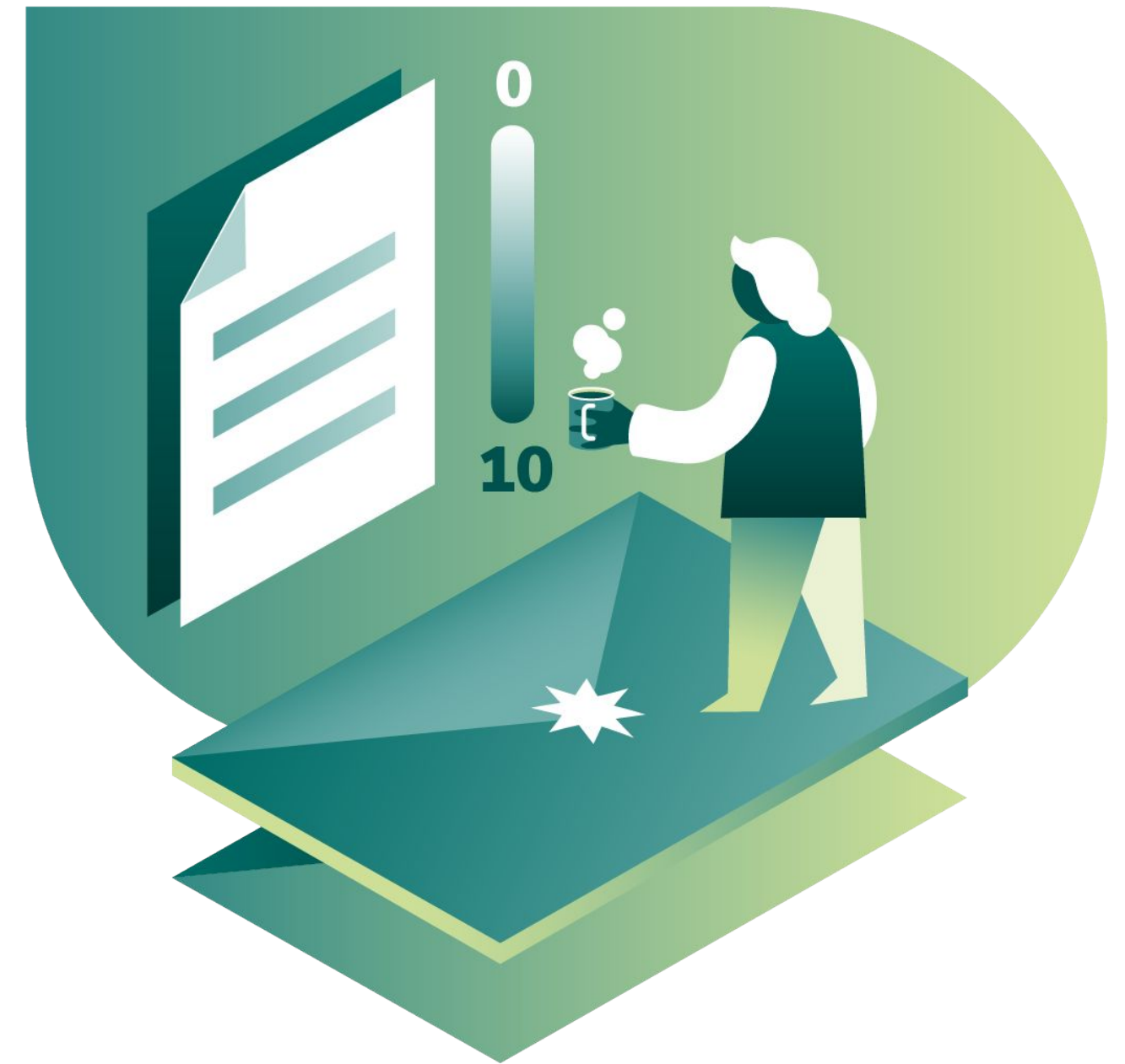
“Consensus decision making is a process where a group agrees on a decision that everyone supports, even if some don't agree with every aspect.”



Deliberation

- ♥ **10** Excellent - Adopt as-is
- 7-9** Good - Minor concerns only
- 5-6** Acceptable - Some issues but workable
- ✕ **3-4** Significant concerns - Needs rework
- 0-2** Unacceptable - Fundamentally flawed

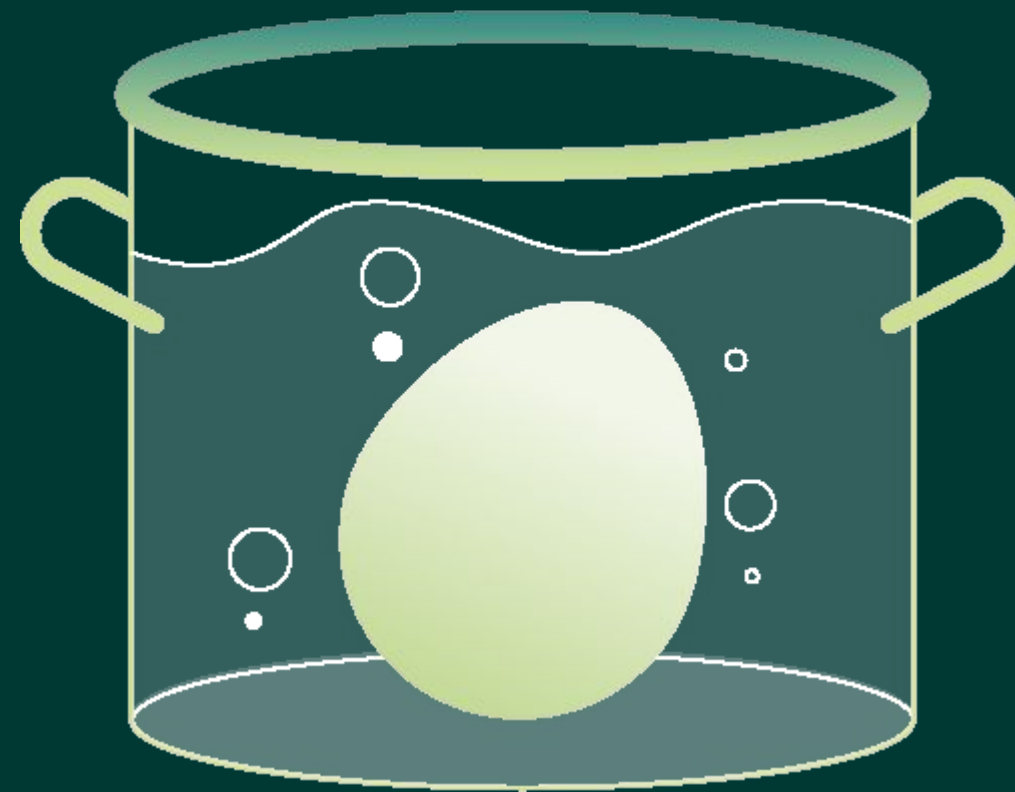
“This isn't voting or selection. It's deliberation. One of the biggest indicators that it's working is watching minds change”



The GSF Assembly Canteen



Scrambled

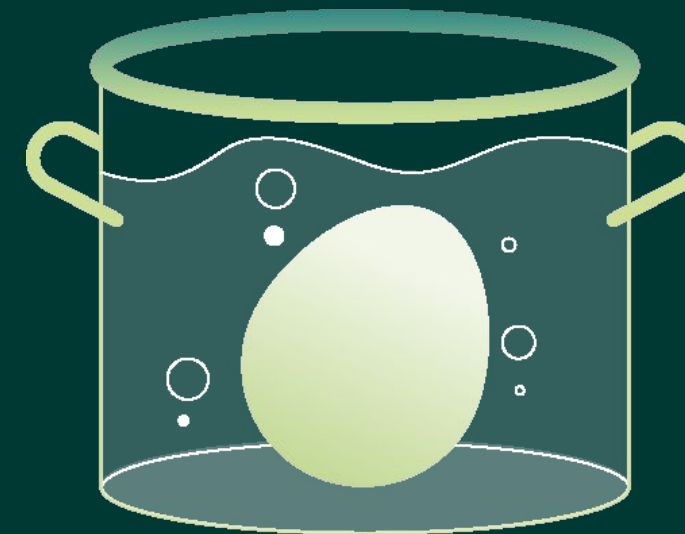


Poached

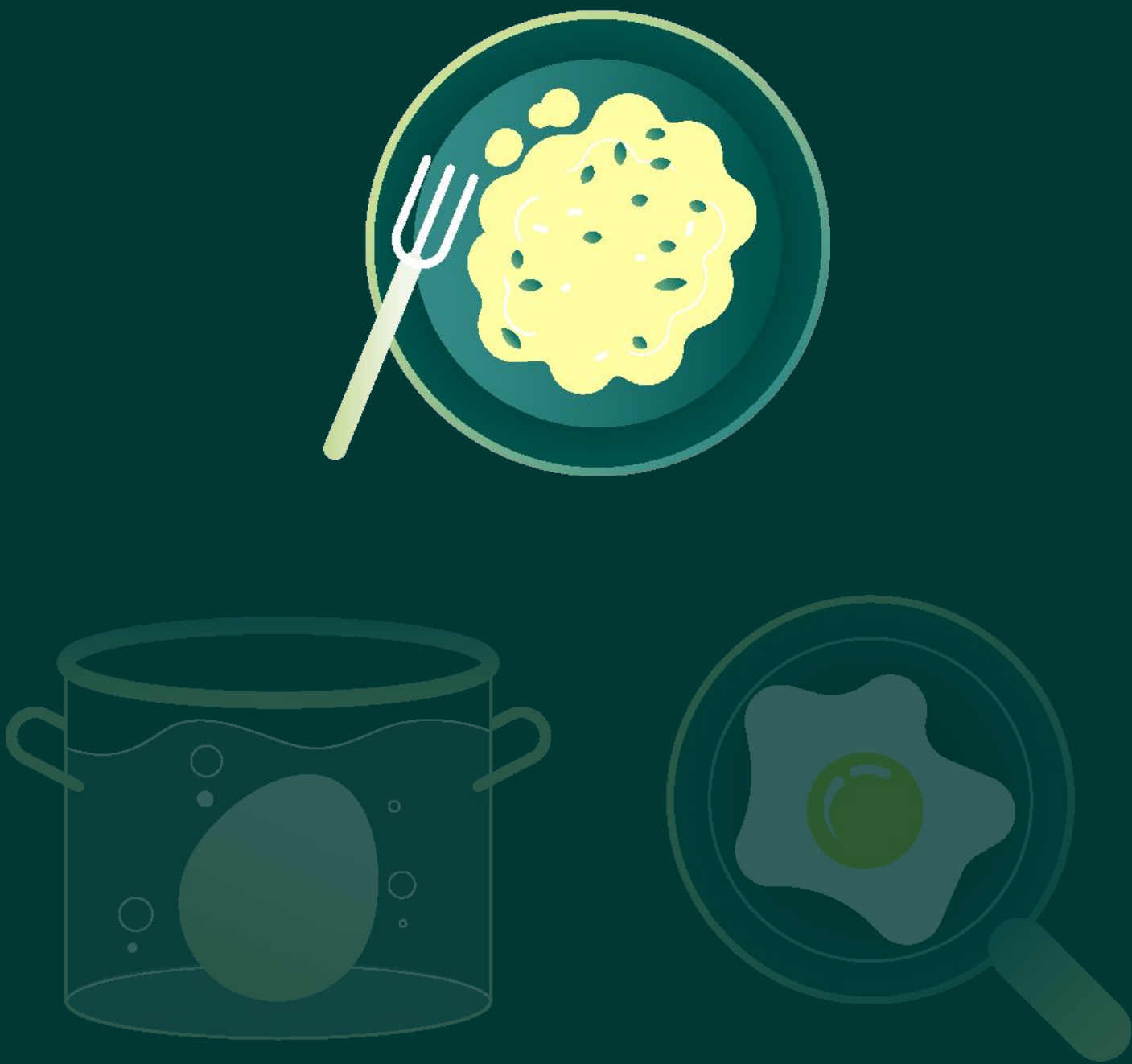


Fried

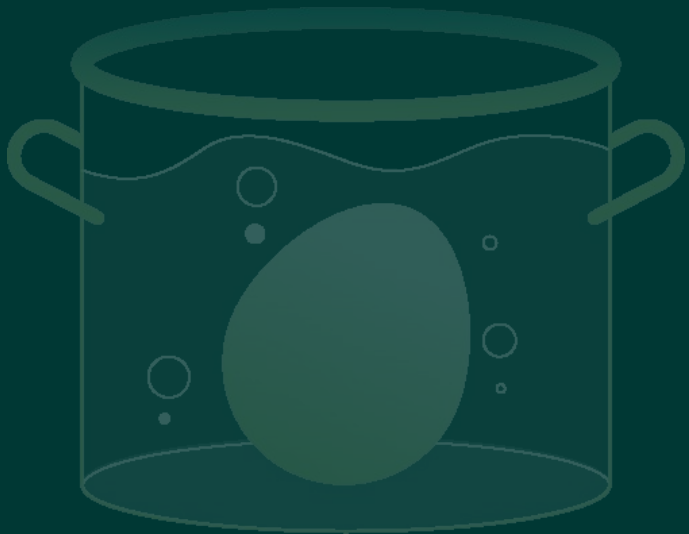
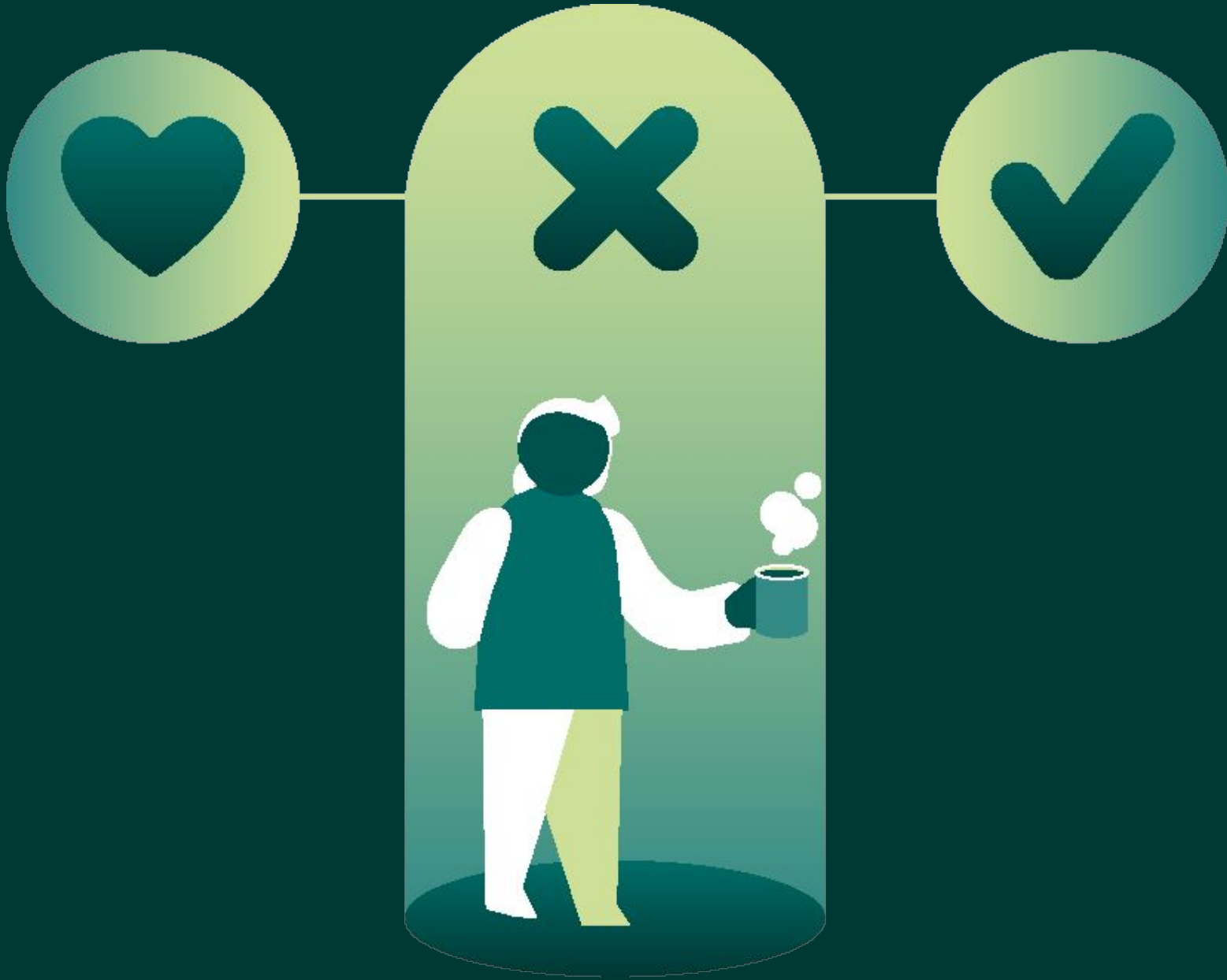
Round 1: Rate each from 0 to 10



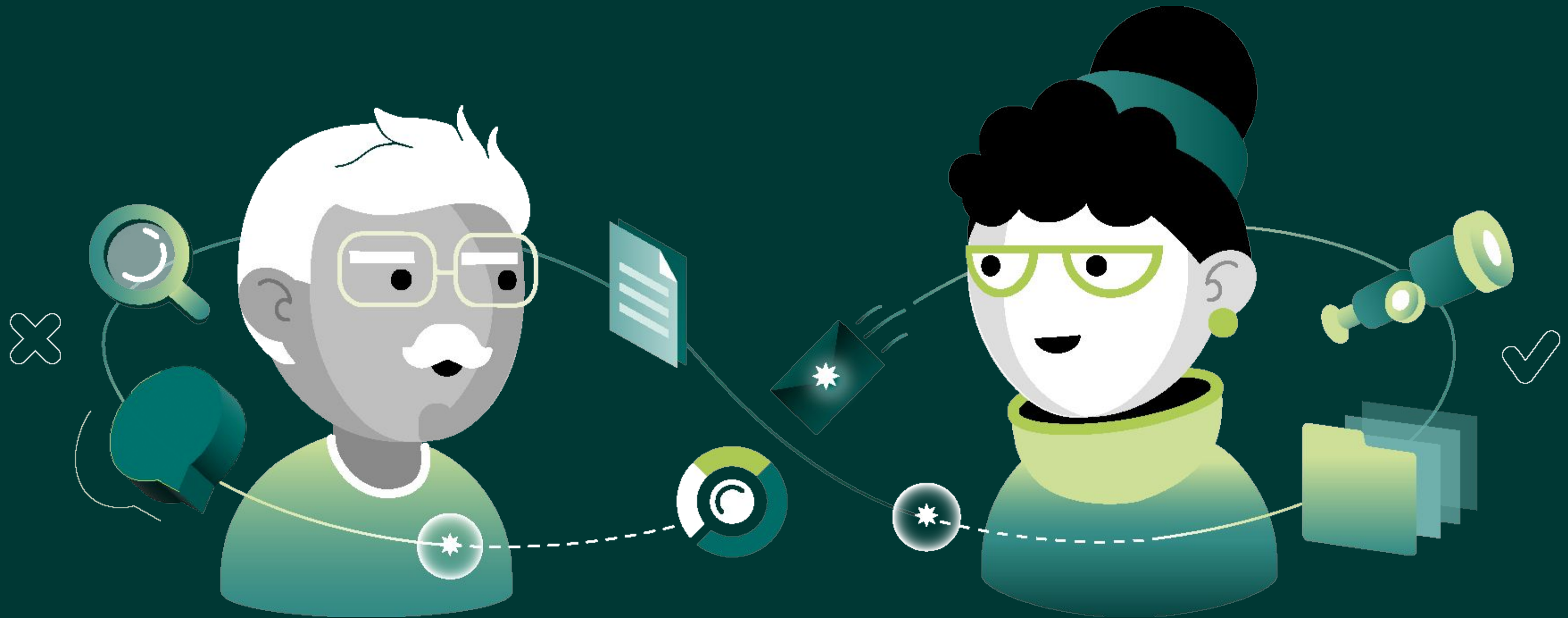
Round 2: Endorse, Consent, Object



Round 3: Objection Resolution



Humans-in-the-Loop



“What first got you into green software or observability?”

Before Monday

Do Now

Complete your participation agreement if a non-member.

It is required to take part.



Then

Monday 27 July the first email arrives.

Respond to a set of questions in your own words.



Your response is the process.

If you go quiet, you have
agreed.